Reminders and announcements

- During lectures you can submit questions to: vevox.app ID: 107-018-952
- You can find answers to questions in the COMP1870 Teams Channel (link in Minerva).
- Note that the deadline for portfolio submissions is 2.00pm, 10 days after the worksheet is posted.
  - We expect and encourage everyone to make their portfolio submissions by the deadline.
  - There is another submission portal you can use for late submissions and resubmissions.

Revision

Name	**Previous statement(s)**	**New statement**	**Justification**
Conjunction Elimination	$j.\ A \wedge B$	A	$\wedge$Elim j
	$j.\ A \wedge B$	B	$\wedge$Elim j
Conjunction Introduction	$j.\ A$ $k.\ B$	$A \wedge B$	$\wedge$Intro j, k
Implication Elimination	$j.\ A \Rightarrow B$ $k.\ A$	B	$\Rightarrow$Elim j, k
Implication Introduction	$\begin{array}{\|l} \hline j.A \\ \vdots \\ k.B \end{array}$	$A \Rightarrow B$	$\Rightarrow$Intro j-k
Disjunction Introduction	$j.\ A$	$A \vee B$	$\vee$Intro j
	$j.\ A$	$B \vee A$	$\vee$Intro j
Disjunction Elimination	$j.\ A \vee B$ $k.\ A \Rightarrow C$ $l.\ B \Rightarrow C$	C	$\vee$Elim j, k, l
Negation Introduction	$j.\ A \Rightarrow (B \wedge (\neg B))$	$\neg A$	$\neg$Intro j
Double Negation	$j.\ \neg(\neg A)$	A	$\neg\neg\ j$
	$j.\ A$	$\neg(\neg A)$	$\neg\neg\ j$

Proofs

Mathematics is characterised by the use of **deductive logic** to **prove** statements. Constructing proofs is an art as much as a science, and is a difficult skill to learn.

There are many statements that mathematicians do not know how to prove:

- **Riemann hypothesis.** Every non-trivial zero of the Riemann zeta function has real part $\frac{1}{2}$.
- **P $\neq$ NP.** There are problems that can be solved in polynomial time by non-deterministic Turing machines but not by deterministic Turing machines.
- **Goldbach's conjecture.** Every even number greater than two can be expressed as the sum of two primes.

Three important proof styles

- **Direct proof.** We want to prove a statement of the form $X \Rightarrow Y$.

  We assume that X is true and then deduce a sequence of statements, ultimately arriving at Y.

- **Contrapositive proof.** We want to prove a statement of the form $X \Rightarrow Y$.

  We give a direct proof of the equivalent statement $(\neg Y) \Rightarrow (\neg X)$.

- **Proof by contradiction.** We want to prove a statement X.

  We prove that $\neg X$ implies something impossible. That is, we give a direct proof of a statement with the structure $(\neg X) \Rightarrow (P \wedge (\neg P))$.

Example of direct proof

The **integers** are the numbers $\ldots, -2, -1, 0, 1, 2, \ldots$.

Proposition 3.3. Let a and b be integers. If a and b are even, then $a + b$ is even.

Example of contrapositive proof

Proposition 3.4. Let z be an integer greater than one. If $2^z - 1$ is prime, then z is prime.

Example of contrapositive proof

Example of proof by contradiction

Theorem 3.5. There are infinitely many prime numbers.

Example of proof by contradiction

Theorem 3.5. There are infinitely many prime numbers.

Inductive proofs

The **positive integers** are the numbers $1, 2, 3, \ldots$. The **non-negative integers** are $0, 1, 2, \ldots$.

- We typically use a **proof by induction** to prove that some statement is true for all positive integers, or for all non-negative integers.
- We start by proving that the property holds for 1 (in the positive integer case) or 0 (in the non-negative integer case). This step is called **checking the base case**.
- We then prove that if the statement is true up to an arbitrary point in the integers, then it is true for the next integer: if it is true for the numbers $1, 2, \ldots, k$, then it is true for $k + 1$, where k is an arbitrary integer.
- If we can accomplish these steps, we have proved the statement is true for all positive (non-negative) integers.

Inductive proofs

An equivalent explanation, using predicate logic.

- We want to prove that $\forall n\ P(n)$ is true, where the domain is the set of positive integers and P is some predicate.
- We check $P(1)$ is true.
- We prove that when k is an arbitrary positive integer,

$$P(1) \wedge P(2) \wedge P(3) \wedge \cdots \wedge P(k) \Rightarrow P(k+1)$$

- Now $P(n)$ is true for all positive integers n.

Inductive assumption

How do we prove

$$P(1) \wedge P(2) \wedge P(3) \wedge \cdots \wedge P(k) \Rightarrow P(k+1) \ ?$$

We assume $P(1) \wedge P(2) \wedge P(3) \wedge \cdots \wedge P(k)$ is true, and use this assumption to derive $P(k+1)$.

In other words, we assume that the statement is true for all the integers $1, 2, \ldots, k$. This is called the **inductive assumption** or **inductive hypothesis**.

We use this assumption to prove that the statement is true for $k+1$.

Example of an inductive proof

Proposition 4.1. Let n be a positive integer. Then

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

Example of an inductive proof

Prime decomposition

Here is an inductive proof using the idea of a **minimal counterexample**.

Theorem 4.4. Any integer greater than 1 is either prime or is equal to a product of prime numbers.

Prime decomposition

Prime decomposition

Inductive proofs in computer science

We frequently use inductive proofs in computer science in these applications:

- analysing running time of algorithms, and
- proving the correctness of algorithms.

Analysing running times

Imagine that an algorithm has running time $T(n)$ when the input size is n.

We know that $T(1) = 1$ and $T(2) = 2$, and $T(n) = T(n-1) + 2T(n-2)$ when $n > 2$.

Analysing running times

Analysing running times

Analysing running times

Weak/strong induction

Sometimes we do not need the full inductive assumption.

If we only use the assumption that $P(k)$ is true, rather than the full assumption that $P_1, P_2, \ldots, P(k)$ are all true, we have used **weak induction**.

Otherwise we have used **strong induction**.